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Complex Training for Power Development: Practical Applications for Program Design

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ABSTRACT

THE SHORT-TERM GAINS IN POWER AND RATE OF FORCE DEVELOPMENT AFTER MAXIMAL OR HIGH-INTENSITY DYNAMIC EXERCISES ARE THOUGHT TO RESULT FROM POSTACTIVATION POTENTIATION (PAP). THE MAJOR FACTORS AFFECTING PAP UTILIZATION ARE THE OPTIMAL INTRACOMPLEX RECOVERY, TRAINING STATUS, AND STRENGTH LEVELS OF THE ATHLETES. STUDIES HAVE SHOWN THAT WITH THE IDEAL COMBINATION OF MODERATELY HIGHLY TRAINED ATHLETES AND ADEQUATE INTRACOMPLEX RECOVERY, IT IS POSSIBLE TO EFFECTIVELY IMPLEMENT COMPLEX TRAINING FOR POWER DEVELOPMENT. THIS PAPER LOOKS TO REVIEW THE CURRENT LITERATURE OF STUDIES INVESTIGATING THE CHRONIC ADAPTATIONS OF PAP IN A TRAINING CYCLE AND RECOMMEND AN EFFECTIVE AND PRACTICAL COMPLEX TRAINING PROGRAM.

INTRODUCTION

Many aspects of a strength and conditioning program center on power development with

complex training being commonly used. Complex training involves maximal or high-intensity dynamic exercises before performing a lighter-resistance ballistic movement with similar biomechanical characteristics (7). Recent research has shown that maximal or high-intensity dynamic exercises (e.g., heavy squats, weighted countermovement jumps and drop jumps) can enhance the rate of force development and jump height of both vertical and horizontal jump performance (5,16,29). This training technique takes advantage of postactivation potentiation (PAP) which is defined as the enhanced neuromuscular condition observed in the skeletal muscle after an initial bout of heavy resistance exercise (27).

The short-term increases in power after maximal or high-intensity dynamic exercises are thought to result from a combination of 2 physiological phenomena. The first theory focuses within the localized muscle where the increased recruitment of high threshold motor units (7,18,24,28) and phosphorylation of myosin regulatory light chains makes the actin and myosin more sensitive to Ca^{2+} released from the sarcoplasmic reticulum (23). This increases the rate of binding of actin and myosin resulting in faster muscle contraction (13). The second theory focuses on the

spinal level where the potentiated muscular state is attributed to an increase in α -motoneuron excitability as reflected by changes in the H-reflex (6,25). The H-reflex is a reflexive neural signal, which when superimposed on voluntary muscle activation, increases the strength of the electrical impulse, thus activating more motor units (6).

PAP has been shown to increase the rate of force development of the affected muscle groups which leads to an increase in acceleration and velocity (26). Both acute and chronic increases in muscular strength and power may be further enhanced by performing an explosive power exercise while the affected muscle groups are in this potentiated state (27).

INTRACOMPLEX RECOVERY

One of the major factors affecting PAP utilization is the optimal intracomplex recovery (i.e., rest interval between maximal or high-intensity dynamic exercise and ballistic exercise) (7,18,24,28). A muscular contraction produces both PAP and fatigue and it is the next balance between these 2 variables that determines whether the subsequent performance response is

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postactivation potentiation; complex training; chronic adaptations

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enhanced, reduced, or unchanged (Figure) (14). During the rest interval, muscle performance may improve if potentiation dominates and fatigue is reduced, decrease if fatigue dominates over potentiation and remain unchanged if both fatigue and potentiation are at similar levels.

Since PAP coexists with fatigue, it is vital to identify the optimal rest interval whereby the muscle has partially recovered from fatigue but is still in a potentiated state (3). Previous reviews in the assessment of the temporal profile of PAP had reported a lack of consensus regarding the optimal intracomplex recovery with the recovery interval ranging from 3 to 10 minutes (19,28). To implement complex training in a training cycle, a shorter recovery interval of ~3–4 minutes would be ideal for strength and conditioning coaches to execute an effective, yet practical training program when performing and incorporating PAP.

TRAINING STATUS AND STRENGTH LEVELS

Other major contributing factors affecting PAP utilization are an athlete's training status, resistance training experience, and strength level

(7,18,24,28). To adhere to the recommended ~3–4 minutes intracomplex recovery, it is recommended to implement complex training in training cycles of moderately highly trained athletes with high relative 1 repetition maximum (1RM) strength levels (training status = club, professional and elite athletes; resistance training experience ≥ 2 years; lower body strength levels ≥ 1.8 relative 1RM; upper body strength levels ≥ 1.4 relative 1RM) (15,18–20). The ability of stronger individuals to express their greatest PAP effect earlier may be explained by the fact that they develop fatigue resistance to heavier loads after a near-maximal effort (1). Given the interplay between strength, fatigue, and potentiation, stronger and experienced individuals may be able to dissipate fatigue quicker after the maximal or high-intensity dynamic exercise because of their greater capacity to resist fatigue and therefore may be able to achieve their maximal PAP response earlier than weaker individuals (1). Stronger individuals may have a higher percentage of type II muscle fibers and therefore likely exhibit greater increases in myosin RLC phosphorylation in response to dynamic

exercise or respond more to increases in the ability to recruit type II muscle fibers resulting in a greater voluntary PAP response (21).

In contrast, moderately trained athletes may incorporate the use of plyometric exercises to induce the effect of PAP for complex training within a training cycle. Plyometric exercises are associated with the preferential recruitment of type II motor units which is one central level mechanism underpinning PAP (6). One study directly compared the effect of a plyometric versus traditional resistance exercise and reported a greater PAP effect after the former (16). Twelve trained volleyball players performed a variety of specific warm-up stimuli (unloaded and loaded countermovement jumps and drop jumps) after baseline measurements on randomized separate occasions. Jump height and maximal power output significantly improved by 2–5% and 2–11%, respectively.

A recent meta-analysis has also shown that a plyometric exercise may produce less fatigue than a loaded traditional resistance exercise as a conditioning stimulus, thus allowing a greater potentiation effect to be achieved and reducing the time necessary to achieve the maximal PAP effect (19). Given the relationship between fatigue and PAP, a plyometric exercise may produce less fatigue than a loaded traditional resistance exercise, thus allowing a greater potentiation effect to be achieved and reducing the intracomplex recovery needed to achieve the maximal PAP effect (19).

CHRONIC ADAPTATIONS OF POSTACTIVATION POTENTIATION IN A TRAINING CYCLE

Only a handful of studies have investigated the effectiveness of complex training within a training cycle of 6–10 weeks (4,8–11,17,22) (Table 1). The majority of studies showed that club and elite level athletes taking part in a periodized complex training cycle showed significant improvements in

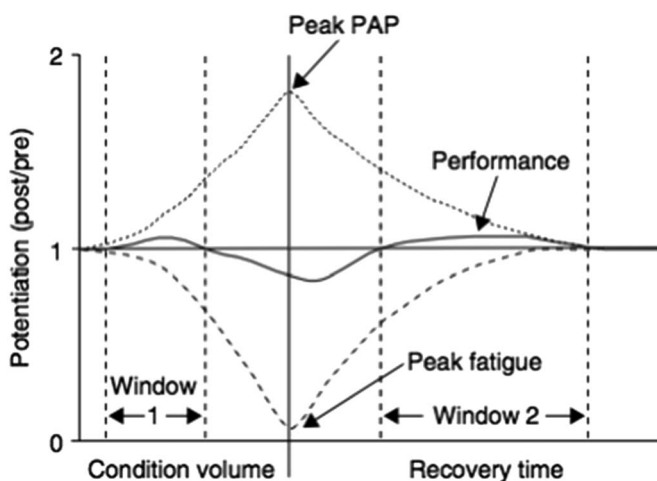


Figure. A model of the relationship between postactivation potentiation (PAP) and fatigue after a previous conditioning stimulus (Adapted from Tillin and Bishop, 2009). PAP can be achieved earlier after a low-volume conditioning stimulus (window 1) or later after a high-volume conditioning stimulus (window 2).

Table 1
Studies investigating chronic adaptations of postactivation potentiation on performance measures

Study	Subjects	Training program			Duration	Performance test	Performance changes
Dodd and Alvar (4)	45 Div II	Complex Training	Plyometric Training	Resistance Training	8 wk	20–60-yd sprint	↑0.27–0.55%
	College baseball players	BS + BJ	BJ	BS		VJ	↑0.98%
		Lunge + DJ	DJ	Lunge		SBJ	↑1.8%
		Split Squat + Split Squat Jumps	Split Squat Jumps	Split Squat		T-agility	↑2.33%
		Notes	Notes				
		Resistance exercises = 4 × 6 at 80–90% 1RM	<10 s between complex pairs				
		Plyometric exercises = 4 × 6 at 0–30% 1RM	3–4 min rest between sets				
			2 d per week				
Juárez et al. (8)	16 Undergraduates	Complex Training	Conventional Training		8 wk	CMJ	VJH ↑ 5.4 cm (Complex)
		2 × 8 BS at 70% 1RM + 5 VJ + 2 × 20 m (week 1–2)	BS (week 1–4)				VJH ↔ (Conventional)
		2 × 6 BS at 75% 1RM + 5 HJ + 2 × 20 m (week 3–4)	VJ + 4 × 20 m (week 5)				
		2 × 4 BS at 80% 1RM + 5 DJ + 2 × 20 m (week 5–6)	HJ + 5 × 20 m (week 6)				
		2 × 4 BS at 85% 1RM + 5 BJ + 2 × 20 m (week 7–8)	DJ + 4 × 20 m (week 7)				
		Notes	BJ + 4 × 20 m (week 8)				
		No rest between complex pairs	Notes				

(continued)

Table 1
(continued)

		3–5 min rest between sets	Resistance exercise in week 1–4 followed by power exercises in week 5–8				
		2 d per week	3–5 min rest between sets				
			2 d per week				
MacDonald et al. (9)	34 M Undergraduates	Complex Training	Plyometric Training	Resistance Training	6 wk	CMJ & SBJ	CMJ pGFF
		Day 1	Day 1	Day 1			↑ 4.1% (Plyometric)
		BS + Lateral Jump	Lateral Jump	BS			↑ 3% (Complex)
		Stiff Leg Deadlift + DJ	DJ	Stiff Leg Deadlift			↔ between Complex, Plyometric & Resistance
		Standing Calf Raise + BJ	BJ (3 × 3–7)	Standing Calf Raise (3 × 3–6 at 75–90% 1RM)			
		<i>Resistance (3 × 3–6 at 75–90% 1RM)</i>	Day 2	Day 2			
		<i>Plyometric (3 × 3–7)</i>	Lateral Jump	Speed Squats			
		Day 2	DJ	Stiff Leg Deadlift			
		Speed Squats + Lateral Jump	BJ	Standing Calf Raise			
		Stiff Leg Deadlift + DJ	<i>Plyometric (3 × 3–6)</i>	<i>Resistance (3 × 3–6 at 45–60% 1RM)</i>			
		Standing Calf Raise + BJ	Notes				

Table 1
(continued)

		<i>Resistance (3 × 3–6 at 45–60% 1RM)</i>	3 min rest between sets	Notes			
		<i>Plyometric (3 × 3–6)</i>		3 min rest between sets			
		Notes					
		3 min rest between exercises					
		3 min rest between sets					
Maio Alves et al. (10)	23 M	Complex Training (G1 & G2)	Resistance	Control	6 wk	SJ	VJH ↑ 12.6% (G1)
	Young elite	6 reps squats at 85% 1RM + 5 m high skipping + 5 m Sprint	BS (week 1–4)	No Resistance or Plyometric training			VJH ↑ 9.63% (G2)
	Portuguese soccer	6 reps calf raise at 90% 1RM + 8 VJ + 3 high ball headers	VJ + 4 × 20 m (week 5)				VJH ↔ between G1 & G2
		6 reps seated leg extension at 80% 1RM + 6 Jumps from seated position + 3 DJ executing soccer heading	HJ + 5 × 20 m (week 6)				
		Notes	DJ + 4 × 20 m (week 7)				
		G1: Train once a week	BJ + 4 × 20 m (week 8)				
		G2: Train twice a week	Notes				
		Rest period: NA	<10 s between complex pairs				
			3–4 min rest between sets				
			2 Days per week				

(continued)

**Table 1
(continued)**

Mihalik et al. (11)	31 (11 M, 20 F)	Complex Training	Compound Training		4 wk	CMJ	VJH ↑ 5.4% (Complex)
	Division 1	Day 1 & 2	Day 1				VJH ↑ 9.1% (Compound)
	Volleyball	BS (3 × 6 at 60% 1RM) + Depth Jump (3 × 6)	BS (6 × 6 at 60% 1RM)				↔ between Complex & Compound
		Single Leg Lunge (3 × 6 at 60% 1RM) + Split Squat Jump (3 × 6)	Single Leg Lunge (6 × 60% at 1RM)				
		Deadlift (6 × 6 at 60% 1RM) + Double Leg Bounds (3 × 6)	Deadlift (6 × 6 at 60% 1RM)				
		Notes	Day 2				
		60 s rest between sets	Depth Jump (3 × 6)				
		2 min rest between exercise	Split Squat Jump (3 × 6)				
			Double Leg Bounds (3 × 6)				
			Notes				
			60 s rest between sets				
			2 min rest between exercise				
Santos and Janeira (17)	25 M	Complex Training		Control	10 wk	SJ	VJH
	14–15 yrs	Day 1	Day 2	No Resistance or Plyometric training		CMJ	↑ 10.5–13% (Complex)

(continued)

Table 1
(continued)

		First plyometric exercise (week 1–2)					
		Second plyometric exercise (week 3–4)					
		Third plyometric exercise (week 5–7)					
		Fourth plyometric exercise (week 8–10)					
Stasinaki et al. (22)	25 M	Complex Training	Compound Training		6 wk	CMJ (Jump & reach method)	VJH ↔ (Complex)
	Young men	Day 1 & 2	Day 1				VJH ↑ 4% (Compound)
	PE students	Leg Press 2 × 6 at 85% 1RM + Leg press throw (2 × 8 at 30% 1RM)	Leg press (4 × 6 at 85% 1RM)				
		Bench Press (2 × 6 at 85% 1RM) + Bench Press Throw (2 × 8 at 30% 1RM)	Bench Press (4 × 6 at 85% 1RM)				
		Smith box squat (4 × 6 at 85% 1RM) + SJ (2 × 8 at 30% 1RM)	Smith box squat (4 × 6 at 85% 1RM)				
		Day 3	Day 2				
		DJ (3 × 8)	Leg Press throw (4 × 8 at 30% 1RM)				
		Notes	Bench Press Throw (4 × 8 at 30% 1RM)				
		3 min rest between complex pairs (e.g., 1a & 1b)	SJ (4 × 8 at 30% 1RM)				
		5 min rest between exercises	Day 3				
			DJ (3 × 8)				

Table 1
(continued)

Notes	3 min between sets	5 min between exercises
↔ = No Significant Difference; BJ = box jump; BS = back squats; CMJ = countermovement jump; COD = change of direction; DJ = drop jumps; F = female; G1 = Group 1; G2 = Group 2; HJ = hurdle jump; M = male; MB = medicine ball; NA = not available; PE = physical education; pGFF = peak ground reaction force; RM = repetition maximum; SBJ = broad jumps; SJ = squat jump; VJ = vertical jump; VJH = vertical jump height.		

lower body power production (i.e., vertical jump height). These studies show how with the ideal combination of moderately highly trained athletes and adequate intracomplex recovery it is possible to effectively implement complex training for power development in a training cycle.

One concern is how to effectively use the rest interval between both the complex pairs and exercise sets (intracomplex recovery: ~3–4 minutes; intercomplex recovery: ~5 minutes). A possible solution is to cater mobility and/or stability drills for the unaffected limbs (i.e., upper body/core corrective exercises for lower body complex exercise sets), with the aim of addressing dysfunctional movement patterns that can cause a decrease in performance and an increase in injuries (2).

Basic movement pattern limitation, due to asymmetrical function of joint mobility and stability, is thought to reduce the effects and benefits of functional training and physical conditioning. If the asymmetrical dysfunction is unattended to, compensatory movement patterns develop during training and the individual creates a dysfunctional movement pattern that is used subconsciously whenever executing an exercise movement (2). This may lead to greater mobility and stability imbalances and deficiencies, which increase the potential for injury (12).

These corrective exercises are implemented during the rest periods (intracomplex recovery) between the conditioning stimulus and ballistic exercise. This may effectively address other injury management concerns of the athletes during training. This prehabilitation training approach can be supplemented into a complex training protocol without unnecessarily extending the total training time. All these are factored into program design to cater to an effective, yet practical training program (see Table 2 for sample program templates tailored

for both a highly and moderately-trained athlete).

SUMMARY

The majority of the studies investigating the effectiveness of complex training within a training cycle showed significant improvements in lower body power production. The major factors affecting PAP utilization are the optimal intracomplex recovery, training status, and strength levels of the athletes. This shows that with the ideal combination of moderately highly trained athletes and adequate intracomplex recovery, it is possible to effectively implement complex training for power development in a training cycle. The key to successfully using PAP into a training cycle is, taking all the above considerations and implementing them in an effective, yet practical training program. The programming of mobility and/or stability drills within the intracomplex and intercomplex recovery interval may be a solution to address other injury management concerns of the athletes during training, without unnecessarily extending the total training time.

PRACTICAL APPLICATION

Guidelines for using PAP within a training program.

1. Ideal subject characteristics
 - Training status = moderately to highly trained athletes
 - Resistance training experience ≥ 2 years
 - Strength levels ≥ 1.8 relative lower body 1RM
 - Strength levels ≥ 1.4 relative upper body 1RM
2. Effective rest interval
 - Intracomplex recovery (between complex pairs) = ~3–4 minutes
 - Intercomplex recovery (between exercise sets) = ~5 minutes
3. Programming mobility and/or stability drills within the intracomplex and intercomplex recovery interval.

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Table 2
Sample complex training programs for a highly trained and moderately trained athlete

No.	Exercises	Tempo	Sets × repetitions	Load	Rest	Notes
Highly trained athlete						
1A	Back Squats	301	4 × 3–5	>85% 1RM	—	
1B	Serratus Wall Slides and/or Isometric Bird-dog Holds	—	4 × 10 + 4 × 5 (3 s hold on each side)	Body mass	—	3–4 min intra-complex recovery
1C	Squat Jump	X	4 × 6–8	0–30% 1RM	~5 min	~5 min intercomplex recovery
Moderately trained athlete						
1A	Drop Jump	X	3 × 6–8	Body mass	—	
1B	Side Lying Extension Rotation and/or Side Bridge	—	4 × 10 + 4 × 30 s (2 rounds per set)	Body mass	—	3–4 min intra-complex recovery
1C	Box Jump	X	4 × 6–8	Body mass	~5 min	~5 min intercomplex recovery



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